

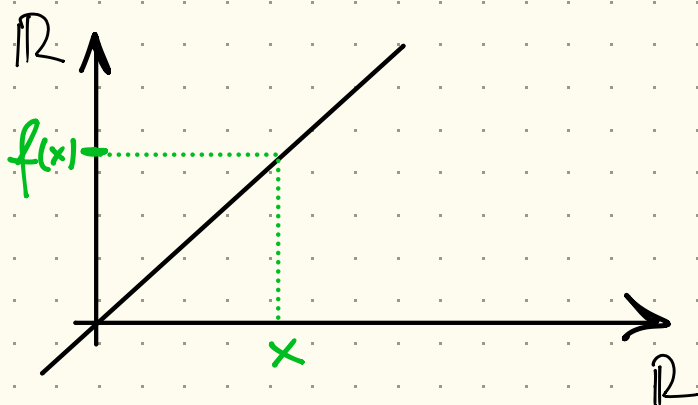
ESERCIZI

$$f: \underbrace{\text{Dom}(f) \subseteq \mathbb{R}}_{\text{DOMINIO}} \rightarrow \underbrace{\mathbb{R}}_{\text{CODOMINIO}}$$

es) $f(x) = x$ $f: \mathbb{R} \rightarrow \mathbb{R}$

NON HA RESTRIZIONI SUL DOMINIO

INPUT = OUTPUT



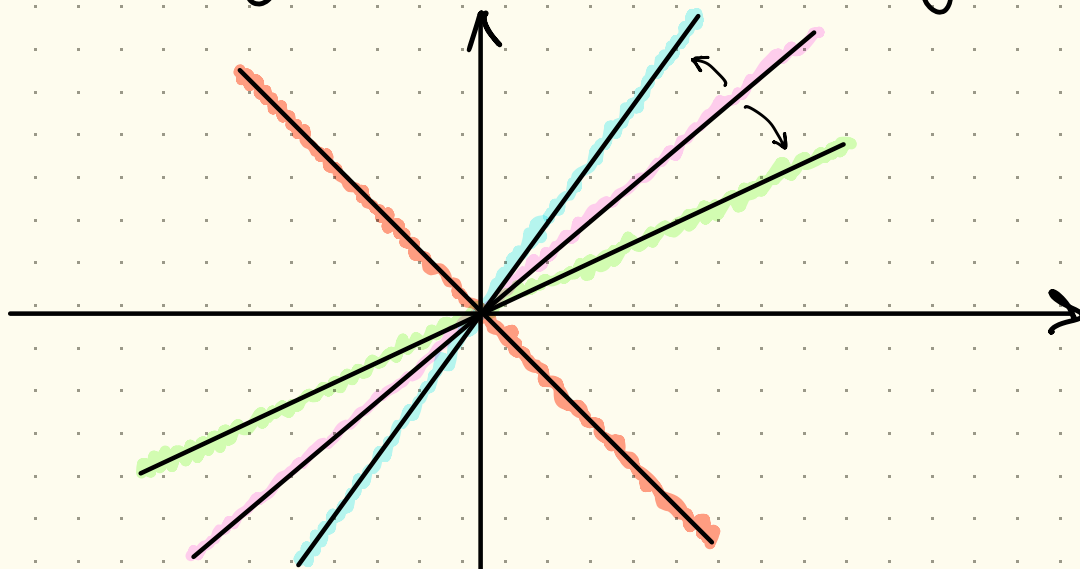
$$\text{Dom}(f) = \mathbb{R} \quad \text{Im}(f) = \mathbb{R}$$

$$y = -x$$

$$y = \frac{1}{2}x$$

$$y = 2x$$

$$y = x$$



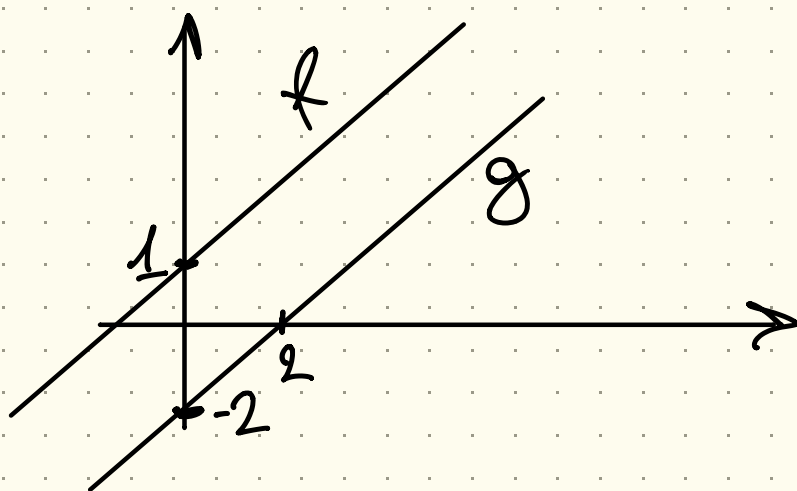
$\forall y \in \mathbb{R} \leftarrow$ SURGETTIVITA' VA VERIFICATA $\forall y \in \mathbb{R}$

$\forall y \in \text{Im}(f) \leftarrow$ INIETTIVITA' VA VERIFICATA SU $\text{Im}(f)$

$y(x)$ $f(x)$

$$y = f(x) = x + 1$$

$\uparrow$ $\uparrow$
 $y = x + 1$ $g(x) = x - 2$



es) $r(x) = ax + b$ $x \in \text{Dom}(r)$

ANALIZZARE DOMINIO IMM. SURG. INIETT. $\forall a, b \in \mathbb{R}$

$r_{a,b}(x) = ax + b$ FORMALI

$a = 1$ $b = 0$

• $r_{1,0}(x) = 1 \cdot x + 0 = x$

• $r_{0,0}(x) = 0$

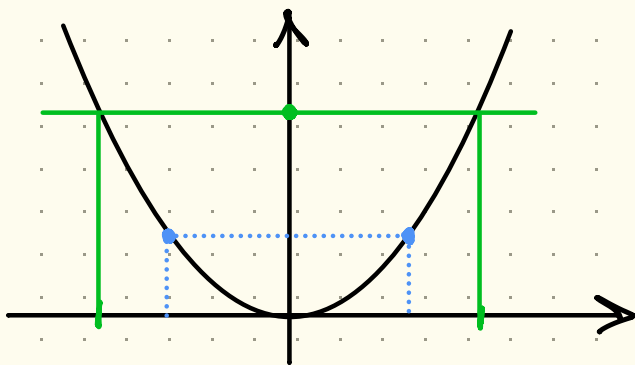
• DOMINIO

$$\text{Dom}(r_{a,b}) = \mathbb{R} \quad \forall a, b \in \mathbb{R}$$

$$\text{Im}(r_{a,b}) = \begin{cases} \mathbb{R} & a \neq 0 \quad \forall b \in \mathbb{R} \\ \{b\} & a = 0 \quad \forall b \in \mathbb{R} \end{cases} \leftarrow \begin{array}{l} \text{É SURGETTIVA} \\ \boxed{\text{SI}} \\ \boxed{\text{NO}} \end{array}$$

$$r_{a,b} = 0 \cdot x + b = b$$

es) $f(x) = x^2 = x \cdot x$



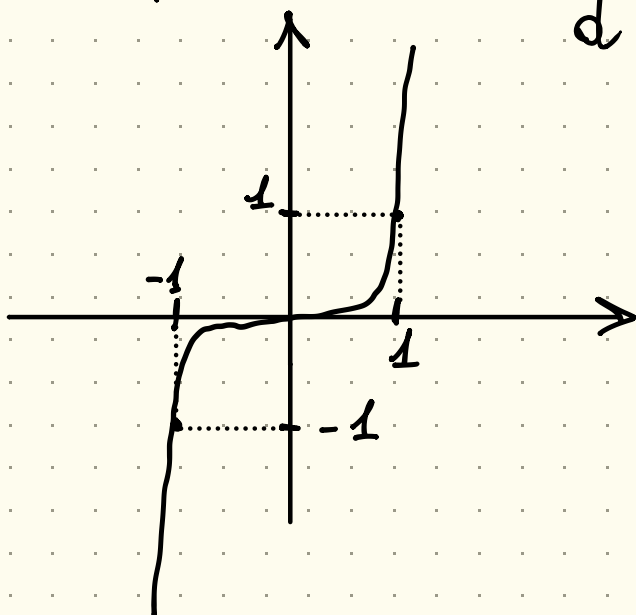
NON É SURG. $\text{Im}(f) \neq \mathbb{R}$

$$\text{Im}(f) = [0, +\infty)$$

É UN
MINIMO

NON É UN
MASSIMO

es) $f(x) = x^d$

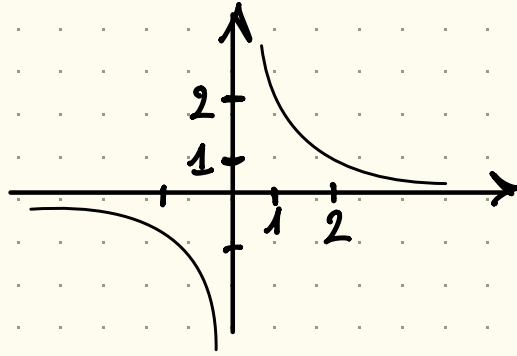


d DISPONIBILI $\text{Im}(f) = \mathbb{R}$

$$\begin{array}{l} y = x^3 \\ y = x^5 \\ y = x^7 \\ \vdots \end{array} \mid \begin{array}{l} \text{Dom}(f) = \mathbb{R} \\ d > 0 \end{array}$$

es) $f(x) = \frac{1}{x}$ $\text{Dom}(f) = \mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, +\infty)$

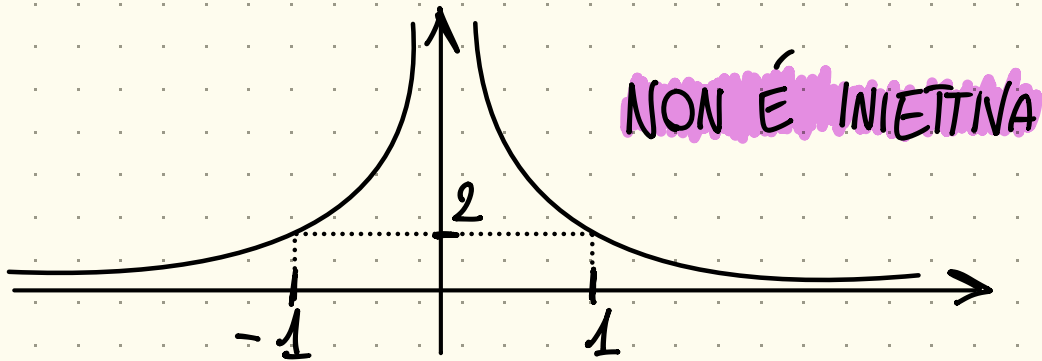
$\underline{x \in \mathbb{R}}$ $f(x) = \frac{1}{x}$ $\boxed{\frac{1}{0} \notin \mathbb{R}}$



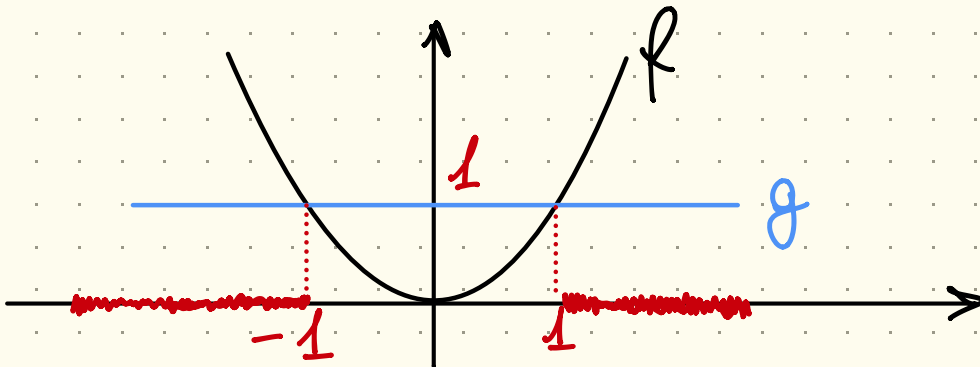
$\text{Im}(f) = \mathbb{R} \setminus \{0\}$

es) $g(x) = \frac{1}{x^2}$ $\text{Dom}(g) = \mathbb{R} \setminus \{0\}$

$\text{Im}(g) = (0, +\infty) \neq \mathbb{R}$



es) $f(x) = x^2$ $g(x) = 1$ $f(x) > g(x)$



$$f(x) > g(x) ? \rightarrow x < -1 \wedge x > +1$$

$$x^2 > 1$$

$$\{x \leq -1\} \vee \{x > +1\}$$

$$x^2 - 1 > 0$$

$$(x-1)(x+1) > 0$$

25) $f(x) = \frac{1}{x}$ $f(x) > g(x)$ $\frac{1}{x} > x$

$$g(x) = x$$

$$x > 0$$

$$1 > x^2$$

$$x^2 < 1$$

$$\rightarrow 0 > x > 1$$

$$x < 0$$

$$1 < x^2$$

$$x^2 > 1$$

$$\rightarrow x < -1$$

